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Mean Contralateral Salivary Gland Dose (Parotid and Submandibular) Accurately Predicts for Patient-Reported Xerostomia After Deintensified Chemoradiation Therapy for Human Papillomavirus–Associated Oropharynx Cancer

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Purpose/Objective(s): To estimate the correlation between different dosimetric indices of the contralateral parotid and submandibular glands with the patient reported severity of dry mouth 6 months post de-intensified chemoradiation therapy.

Materials/Methods: Forty-three patients were treated on a prospective multi-institutional phase II study (ClinicalTrials.gov, NCT01530997) assessing the efficacy of de-intensified chemoradiation therapy in patients with favorable risk, HPV-associated oropharyngeal squamous cell carcinoma. All patients received 60 Gy intensity modulated radiation therapy with concurrent weekly intravenous cisplatinum (30 mg/m^2). The protocol specified dosimetric goals for the contralateral parotid and submandibular glands were mean dose < 26 and < 35, respectively. All patients reported severity of their dry mouth (pre- and post-treatment) using the patient reported outcome version of the CTCAE (PRO-CTCAE): i.e. none/mild/moderate/severe/very severe. We correlated individual patient dosimetric data (e.g. Dmean and various Vxs) from the contralateral parotid and submandibular glands (separate and combined) to changes in their self-reported PRO-CTCA severity of dry mouth (baseline to 6 month post-treatment). A change in severity (from baseline) of >/= 2 was considered clinically meaningful. The ability of different dosimetric indices to accurately predict for patient outcomes was assessed through the area under to Receiver Operating Characteristic curve (ROC) and odds ratios (OR).

Results: The combined contralateral glands (parotid and submandibular) had better AUC values than the individual glands. For the combined glands, various Vx values (ranging from V16-V38) and the mean dose had the highest AUC's (80% to 83%). Among these, the optimal predictor appears to be V18 with an AUC of 83%, threshold of 57%, and hazard ratio of 5.1 (95% CI of 1.3-20.2). For the contralateral parotid V18 to V22 had the highest AUC's (78% to 79%) and the Dmean was not as strong a predictor (AUC 76%, threshold of 23Gy). Vxs of V27 to V36 (AUCs 78% to 79%) of the contralateral submandibular glands were better predictors of xerostomia than Dmean (AUC 71%).

Conclusion: The standard dosimetric objectives of mean dose < 26 Gy for the contralateral parotid and < 35 Gy for the contralateral submandibular gland may not be the optimal indices for sparing salivary gland function in patients with HPV-associated oropharyngeal cancer who receive de-intensified chemoradiation therapy. The rate of patient reported xerostomia following de-intensified chemoradiation therapy appears to be best correlated with the V18 of the combined contralateral parotid and submandibular glands.

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Comparison of Chemoradiation Versus Radiation Alone for Elderly Patients With Locally Advanced Head and Neck Squamous Cell Carcinoma: An Analysis of the National Cancer Data Base

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Purpose/Objective(s): Controversy exists on the benefit of chemotherapy to radiation (RT) for the management of head and neck cancers in the elderly, in part due to comorbidities, decreased functional status, and competing risks of death. The purpose of this study was to determine if elderly patients (age $\geq$ 65 years) with locally advanced head and neck squamous cell carcinoma (HNSCC) have improved overall survival (OS) from chemoradiation therapy (CRT) compared to RT alone. We hypothesized that chemoradiation would be associated with better survival for elderly patients compared to RT alone.

Materials/Methods: Using the National Cancer Database, we identified 21,408 elderly patients with non-metastatic AJCC Stage III or IV HNSCC treated definitively with CRT (15,386 patients; 71.9%) or RT alone (6,022 patients; 28.1%) between 1998 and 2011. Three-year survival rates were estimated using the Kaplan-Meier method. Crude and adjusted hazard ratios (HR) with 95% confidence intervals (CI) were computed using Cox regression modeling. P values of <0.05 were considered statistically significant.

Results: Primary sites included hypopharynx (3,032; 14.2%), larynx (7,326; 34.2%), and oropharynx (11,050; 51.6%). Median follow-up was 24.0 and 44.8 months for all and surviving patients respectively. The median age of patients receiving CRT vs. RT was 70 years vs. 74 years, respectively ($p < 0.0001$). More patients in the CRT group had stage IV disease relative to the RT group (65.0% vs. 52.4%; $p < 0.0001$). Hypopharynx, larynx, and oropharynx primary sites accounted for 14.3%, 32.5%, and 53.2% of the CRT group and 13.7%, 38.6%, and 47.7% of the RT group, respectively ($p < 0.0001$). Elderly patients receiving CRT had significantly improved OS compared with RT alone [HR 0.60 (95% CI: 0.58-0.62)]. 3 year OS was 51.7% in the CRT group vs. 32.4% in the RT group ($p < 0.0001$). On multivariate analysis (MVA) adjusting for age, stage, and primary site, elderly patients receiving CRT had significantly improved OS compared to RT alone [HR 0.67 (95% CI: 0.63 - 0.68)]. This OS benefit persisted when adjusting for race, insurance status, Charlson Deyo comorbidity score, facility type, income, tumor grade, and tumor size in addition to age, stage, and primary site [HR 0.64 (95% CI: 0.59 - 0.69)]. In a subset analysis of patients 70 years and older (12,773 patients), CRT was also associated with improved OS over RT alone on MVA [HR 0.64 (95% CI: 0.58-0.71)].

Conclusion: In this large cohort of elderly patients from a national patient registry with locally advanced HNSCC, receipt of CRT was associated with significantly improved OS over RT alone.

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Management of Large Choroidal Melanoma With I-125 Plaque Brachytherapy: A Case Series of 39 Patients

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Purpose/Objective(s): Enucleation is generally advised when melanoma is >16 mm in diameter and >10 mm in thickness, as was performed in the